

ARTEM MAKLAKOV

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EDUCATION

Moscow Institute of Physics and Technology (MIPT, PhysTech)

2023 - Present

B.S. in Applied Mathematics and Computer Science, expected June 2027

- Cumulative GPA: 4.83/5.00

WORK EXPERIENCE

Huawei Technologies - Research & Development

Software Engineer Intern, Compilers team

July 2025 - Present

- Implemented LLVM backend optimizations for a proprietary architecture (Profile-Guided Optimizations, Link-Time Optimizations, instruction selection/scheduling).

Software Engineer Intern (Summer), AI for Code team

August 2024

- Research on Retrieval-Augmented Generation's impact on code generation

Flow Order Book Exchange (FOB) - industry project mentored by Sberbank

Quant Research / Developer

January 2026 - Present

- Developing a multi-asset LOB→FOB translator that converts external order book liquidity into portfolio flow orders for a batch-clearing exchange
- Extending the model from single-asset execution to multi-asset portfolio demand and matching
- Designing batched execution and clearing logic under inventory, collateral and risk constraints

OPEN SOURCE

LLVM Upstream Contribution

- Merged contribution to LLVM upstream
- Worked with LLVM maintainers during the review process

AWARDS

All-Russian Olympiad of schoolchildren (Physics)

April 2022

The final stage prize-winner

International Experimental Physics Olympiad (IEPHO)

November 2019

Two times bronze medal

November 2020

Level I National Olympiads

2020-2023

Multiple winner in physics and mathematics

CERTIFICATION

IELTS Academic - Overall Band 8.0/9 (CEFR C1)

October 2025

SKILLS

- | | | | |
|----------|----------------------|--------|---------|
| • C/C++ | • LLVM, MLIR | • Go | • CUDA |
| • Python | • Assembly x86 & ARM | • Rust | • Kafka |

PROJECTS

“57Lang” programming language with x86 compiler

December 2023 - May 2024

Tools: C/C++, ASM, Graphviz

- Esoteric programming language with three stages of code processing:
- Frontend generates AST-tree from code
- Middle-end optimizes AST-tree
- Backend translates AST-tree to intermediate representation, which is used by compiler for ELF-file generation

LLVM Pass

April 2025

Tools: C/C++, Graphviz, LLVM

- During compilation stage generates a DOT graph of the instruction flow.
- Instruments the code with logging functions to collect dynamic runtime data, such as: execution results, instruction execution counts.

Mandelbrot set generation

March 2024

Tools: C/C++, ASM, SFML

- Implemented SIMD optimized Mandelbrot set generation and compared performance against scalar implementation

EXTRACURRICULAR

- Teaching assistant at the course of System Programming & Compiler Technologies
- Organized and proctored mathematics/physics olympiads for high-school students
- Volunteer at university events
- Hockey player